Contraception & Abortion Pharmacology

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Learning Objectives

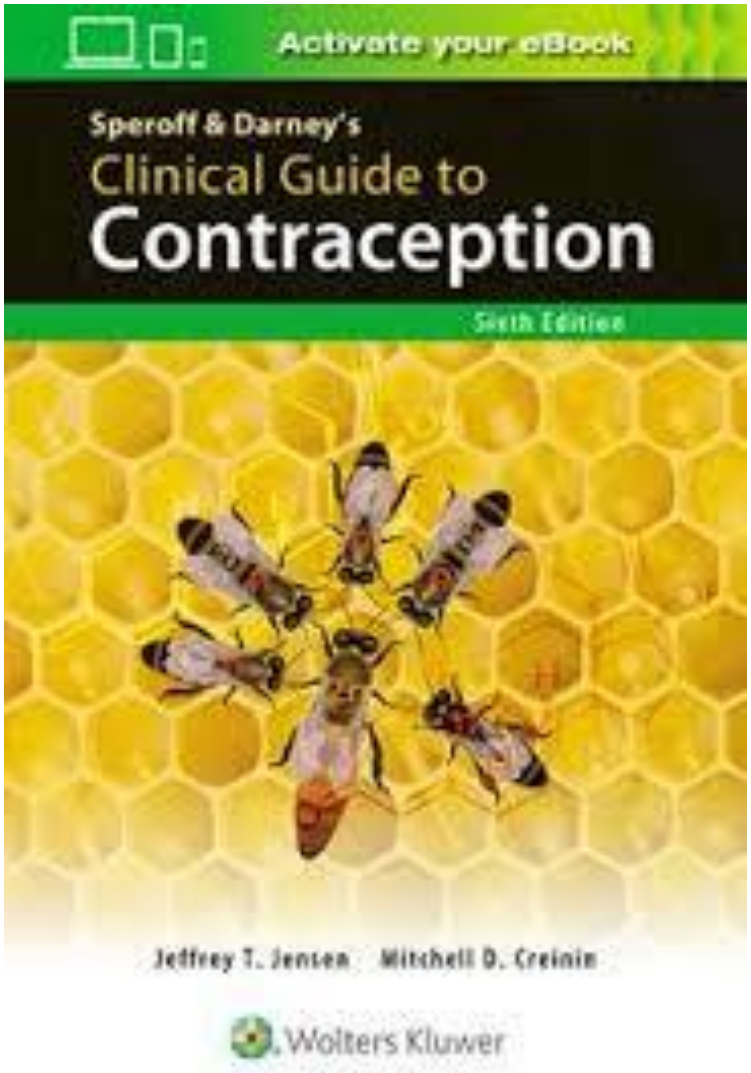
By exam time, you should be able to:

1. Compare and contrast the mechanisms of action, signature adverse effects, and contraindications of combined hormonal (estrogen-progestin) contraceptives.
2. Compare and contrast the different modes of contraceptive delivery (oral, injectable, implantable, intrauterine, vaginal ring, transdermal patch).
3. Describe the rationale for the use of progestin-only contraceptives and long-acting progestins.
4. Classify the agents used for emergency (postcoital) contraception by their MOA and appropriate time frame for use.
5. Describe the mechanisms of action and signature adverse effects of abortion medications.

Disclosures:

None



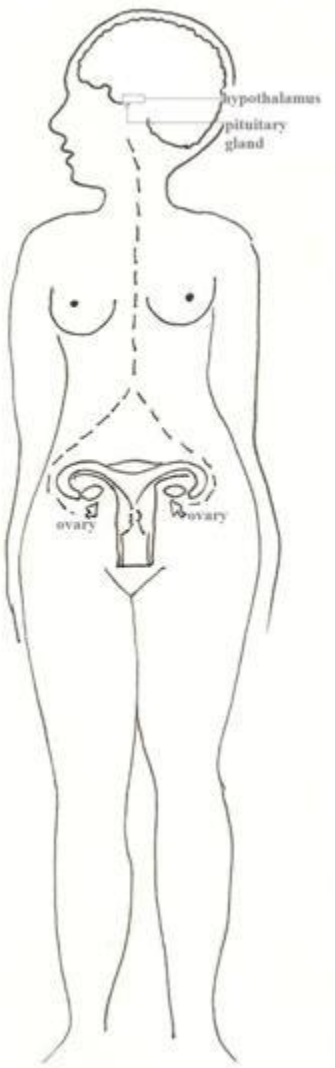


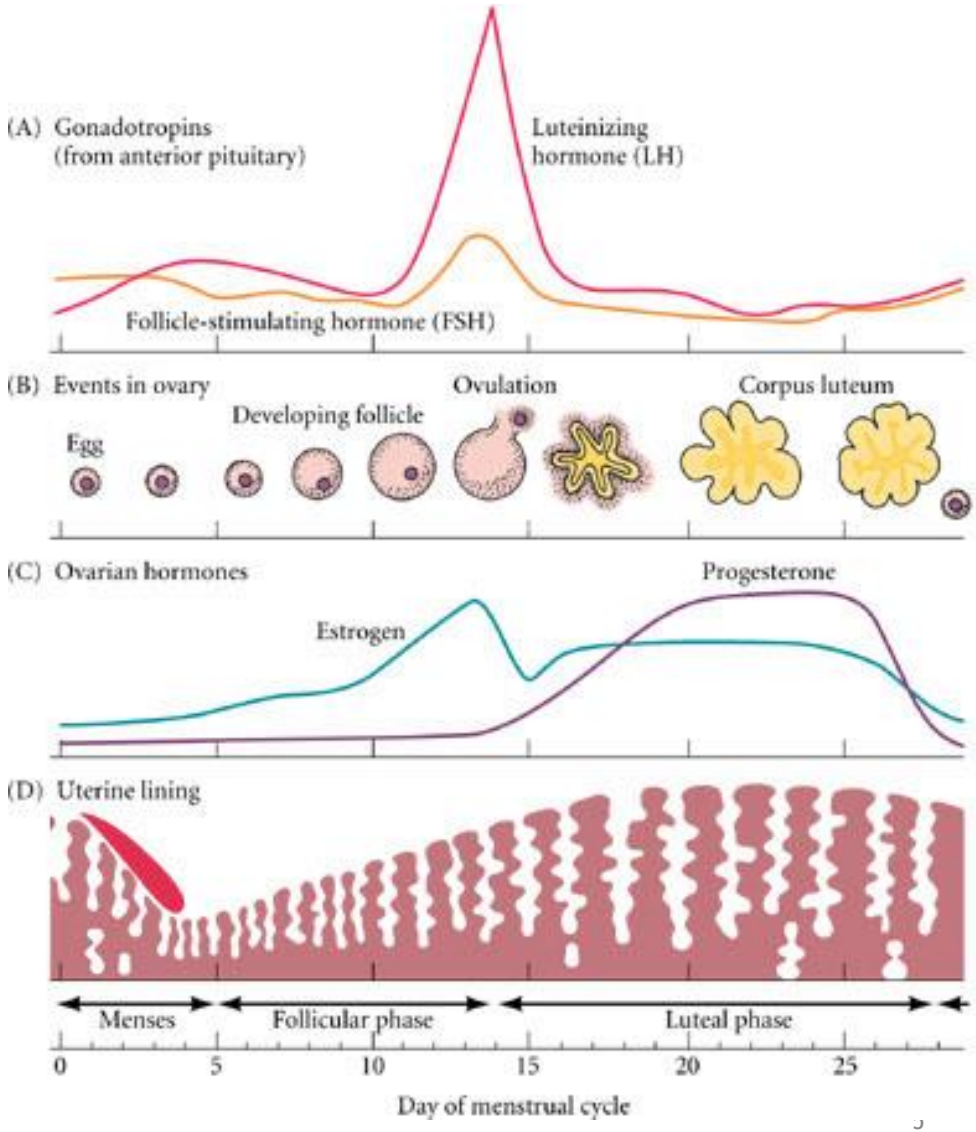
With gratitude to Dr. Lisa Perriera

The Normal Menstrual Cycle

Implant
Depo →
CHCs

The IUDs
POPs →





Contraception at a glance

HOW WELL DOES BIRTH CONTROL WORK?

What is your chance of getting pregnant?

Really, really well



	The Implant (Nexplanon)	IUD (Skyla)	IUD (Mirena)	IUD (ParaGard)	Sterilization, for men and women
Works, hassle-free, for up to...	3 years	3 years	5 years	12 years	Forever

Less than 1 in 100 women

Okay



	The Pill	The Patch	The Ring	The Shot (Depo-Provera)
For it to work best, use it...	Every. Single. Day.	Every week	Every month	Every 3 months

6-9 in 100 women, depending on method

Not so well



	Withdrawal	Diaphragm	Fertility Awareness	Condoms, for men and women

For each of these methods to work, you or your partner have to use it every single time you have sex.

12-24 in 100 women, depending on method

FYI, without birth control, over 90 in 100 young women get pregnant in a year.

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3) What are the components of hormonal contraceptives?

Hormonal contraception

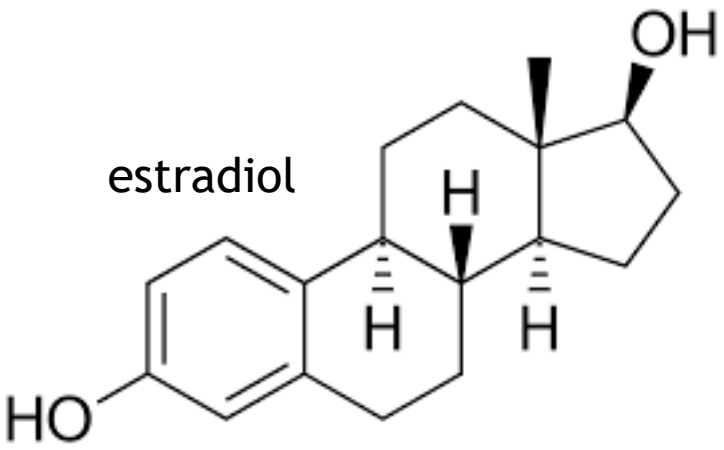
Combined estrogen + progestin
Progestin only

Estrogens

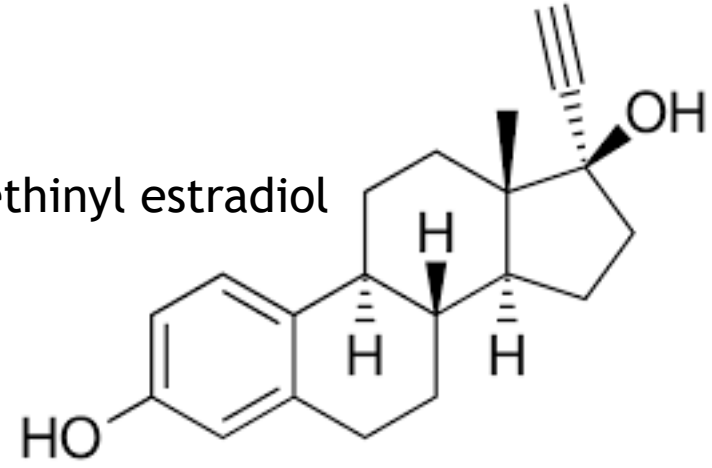
Synthetic Estrogens

- Mestranol → EE: 30% loss in conversion
- **Ethinyl Estradiol (EE)**
  - Bulky side-chain at 17 carbon (ethinyl group) increases binding affinity
- Synthetic Estradiol derivatives
  - Estradiol valerate: pill and injectible
  - Estradiol cypionate- injectable
- Estetrol

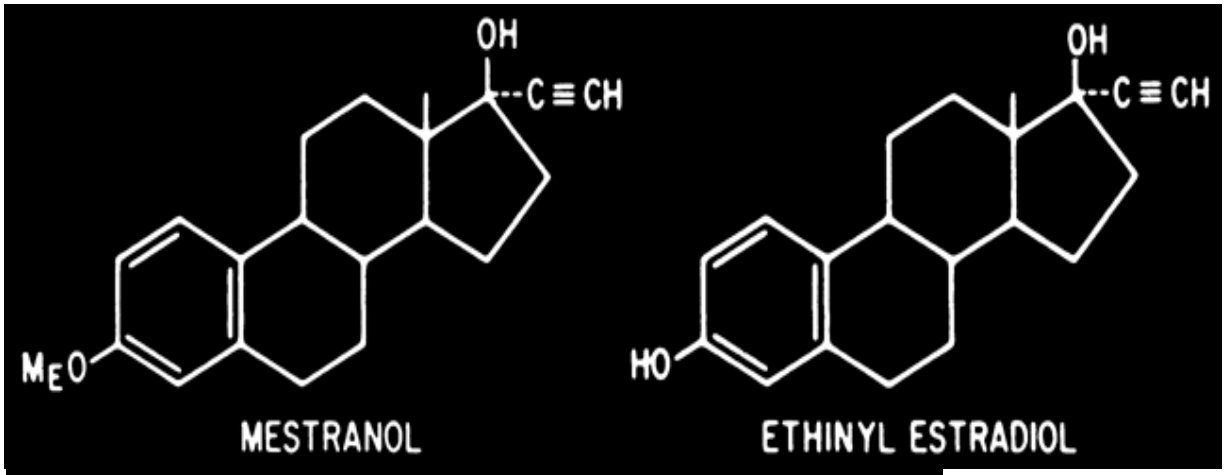
estradiol

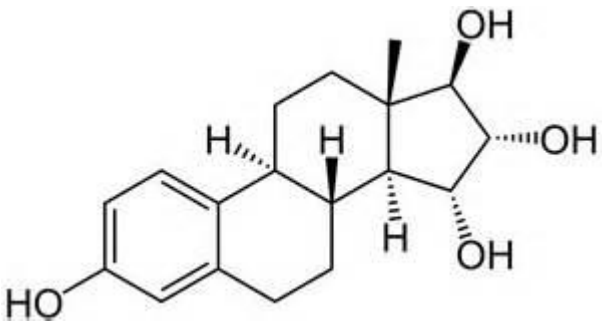


ethinyl estradiol

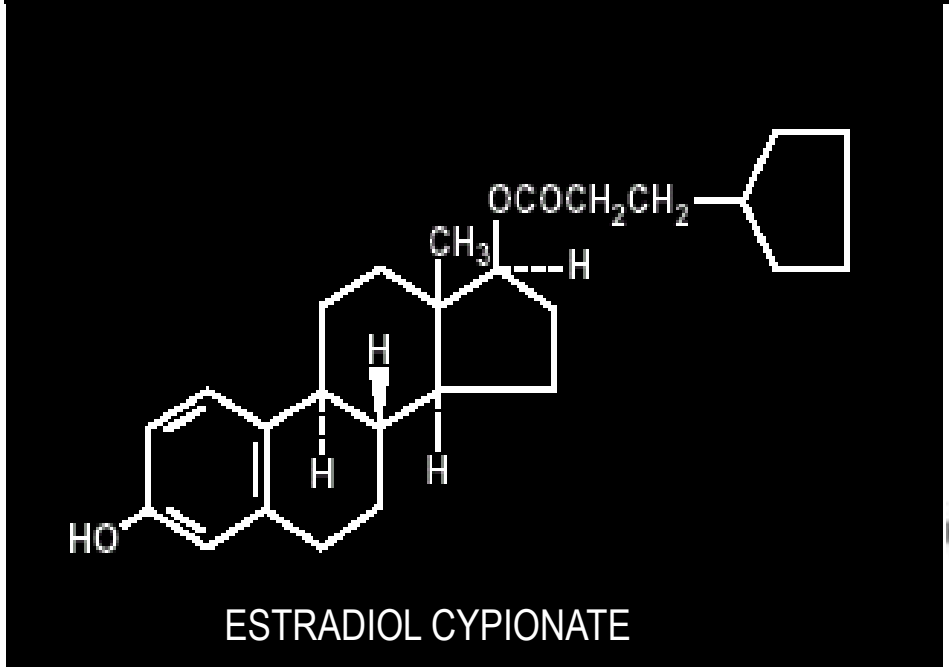


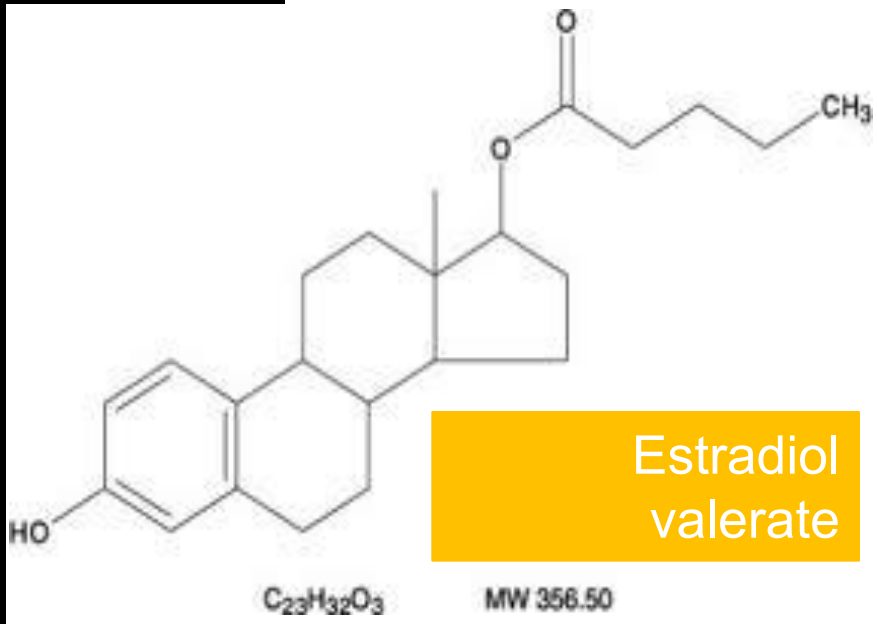
Synthetic Estrogen





Estetrol (E4)



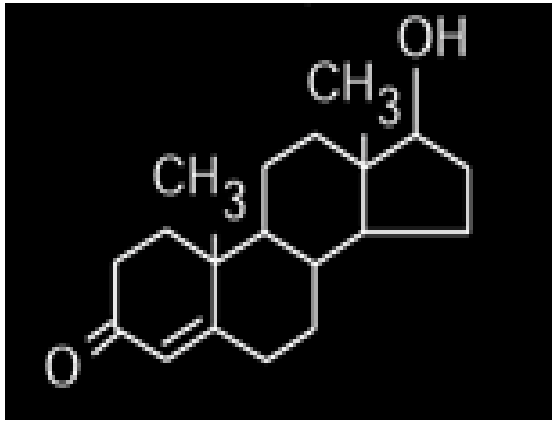


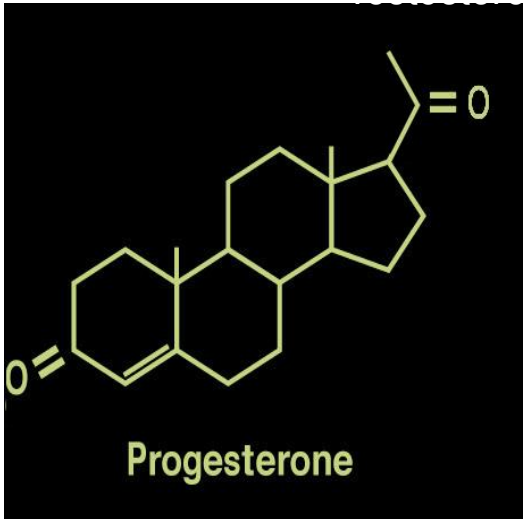
Estradiol valerate

Progestins (synthetic progesterone)

Synthetic progestins

- Testosterone derivatives (19 C)
  - Estranes
  - Gonanes
- Progesterone derivatives (21 C)
  - Pregnanes
  - 19-Norprogesterone
- Spironolactones





Biologically active forms of the estrane progestins

Progestin	*Active Form*
Norethindrone	Norethindrone
Norethindrone Acetate	Norethindrone
Ethynodiol Diacetate	Norethindrone
Norethynodrel	Norethindrone
Lynestrenol	Norethindrone

Components of oral contraceptive pills

Stanczyk FZ et al. *Contraception*. 1990; 42: 67-96

Biologically active forms of the gonane progestins

Progestin	*Active Form*
Levonorgestrel	Levonorgestrel
Gestodene	Gestodene
Desogestrel →	3-keto-desogestrel
Norgestimate →	Levonorgestrel Levonorgestrel-3-oxime Levonorgestrel-17-acetate Other compounds

Component of pills, ring, patch, implant, iud

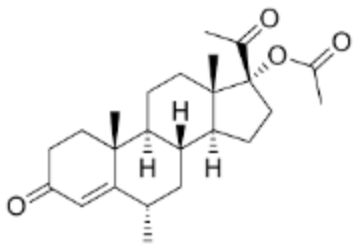
Stanczyk FZ et al. *Contraception*. 1990; 42: 67-96; Stanczyk FZ et al. *Contraception*. 1997; 55: 273-282

Progesterone derived progestins

Pregnane

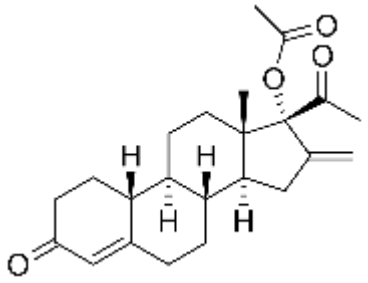
Medroxyprogesterone Acetate

- Derived from 17 α – hydroxyl-progesterone
- Component of contraceptive injection
- Available in oral form
  - Typically used for abnormal uterine bleeding



19-Norprogesterone
Segesterone Acetate

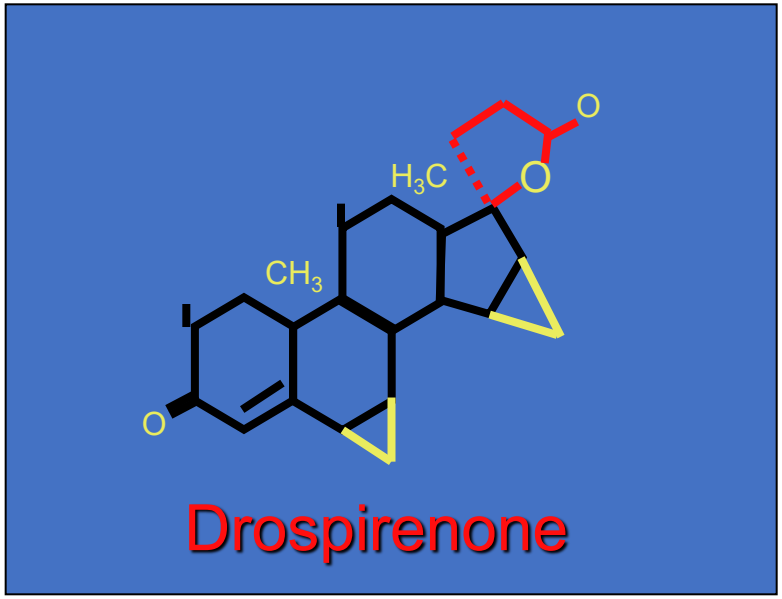
- Progestin component of the year long vaginal ring



Drospirenone

Derived from
17α-spirolactone

Progestin component of
oral contraceptive pills



4) How does contraception work?

Contraceptives can be:

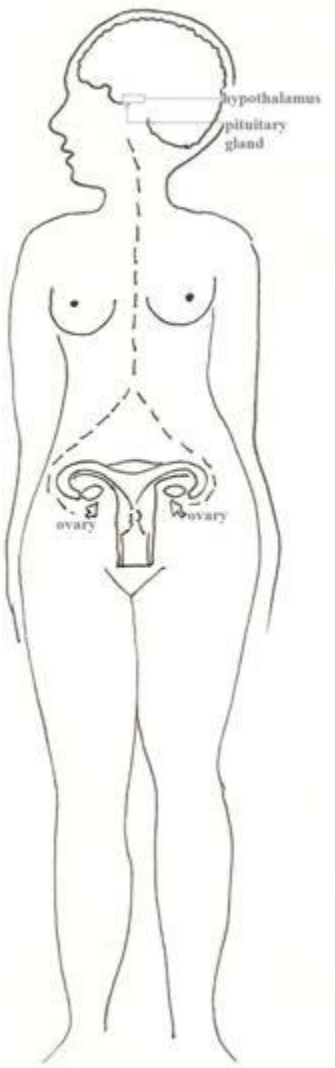
Combined estrogen + progestin
Progestin only
Non-hormonal

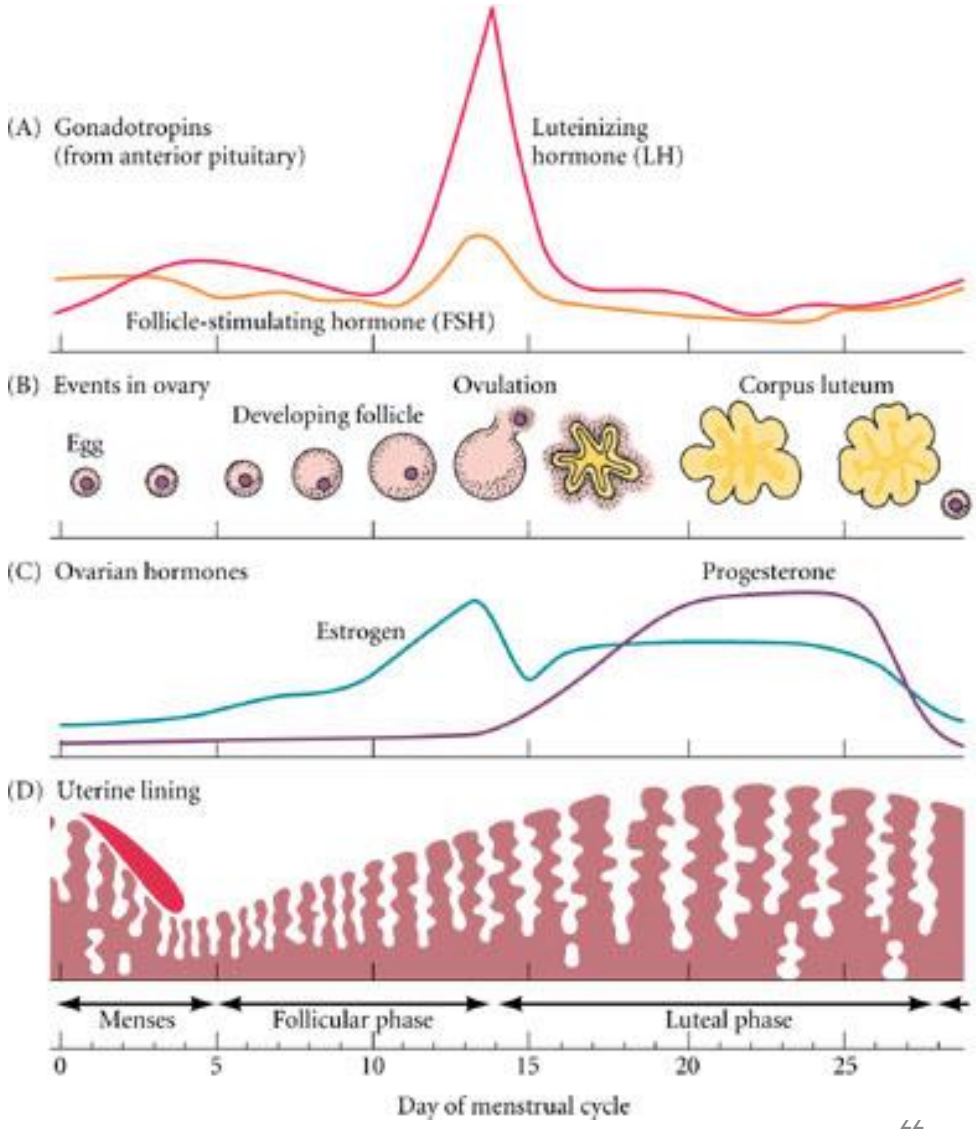
How do contraceptives work?

Combined estrogen + progestin	Ovulation inhibitors
Progestin only	- Cervical mucus thickeners - Ovulation inhibitors
Non-hormonal	- Spermicidal - Barrier

The Normal Menstrual Cycle

Implant
Depo
CHCs





Ovulation Inhibitors



Vaginal Rings





Combined Oral Contraceptives





Contraceptive Patches



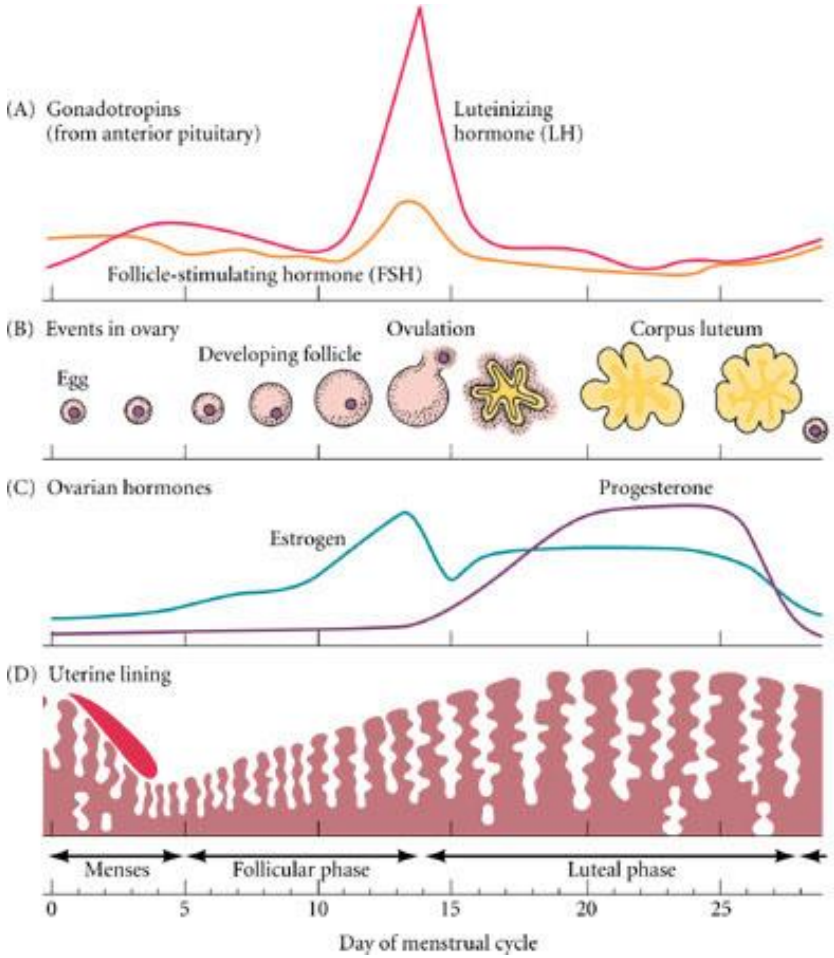
Contraceptive Injection



Contraceptive Implant

5) How do the ovulation inhibitors work?

- Inhibition of gonadotropin secretion
- Progestins
  - suppress LH release
  - Atrophy of endometrial glands
    - Reduces endometrial growth
- Estrogen
  - inhibits FSH release
    - Suppresses follicle development
  - Increases progesterone receptor concentration
    - Allows for lower dose of progestin
  - Stabilizes endometrium



6) Combined hormonal contraception

HOW WELL DOES BIRTH CONTROL WORK?

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Works, hassle-free, for up to...	3 years	3 years	5 years	12 years	Forever

Less than 1 in 100 women

Okay



	The Pill	The Patch	The Ring	The Shot (Depo-Provera)
For it to work best, use it...	[illegible]y. Single. Day.	Every week	Every m[illegible]	Every 3 months

6-9 in 100 women, depending on method

Not so well



	Withdrawal	Diaphragm	Fertility Awareness	Condoms, for men and women

For each of these methods to work, you or your partner have to use it every single time you have sex.

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FYI, without birth control, over 90 in 100 young women get pregnant in a year.

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Combined Hormonal Contraceptive: Delivery Modalities

- **Oral contraceptives**
  - Monophasic vs triphasic vs extended cycle
  - At least 80 different brands



- **Transdermal patch**
  - Two products
    - Norelgestromin /Ethinyl Estradiol (Xulane)
    - Levonorgestrel /Ethinyl Estradiol (Twirla)



- **Vaginal rings**
  - Monthly ring (NuvaRing)
  - Yearly ring (Annovera)



Combined Oral Contraceptives

- *All* combined oral contraceptives (COCs) contain:
  - **A progestin**
  - **An estrogen component**
    - Most with Ethinyl estradiol
    - One with estradiol valerate
    - One with Estetrol
- Daily use
- Preferably the same time every day
- First pass metabolism: absorbed through GI tract, metabolized hepatically
- Failure rate
  - Typical use 7%
  - Perfect use <1%



Transdermal Patches

Norelgestromin patch

- 20 mcg EE + 150 mcg norelgestromin daily (Xulane)
- One patch per week x 3 weeks, then one week off
- Continuous delivery to systemic circulation
  - Area "under the curve" is 60% higher than in a 35 mcg pill
  - Possible increase in estrogen side effects such as VTE
- **Less effective if body weight >198 lbs (90kg)**
- Failure rates
  - Typical: 7%
  - Perfect: 0.3%



Levonorgestrel patch

- Approved February 2020
- 30mcg EE + 120 mcg levonorgestrel daily (Twirla)
- One patch per week x 3 weeks, then one week off
- **Not approved for BMI > 30**
  - Lower efficacy
  - Higher risk of DVT
- Failure rates
  - 4.3% with Bmi <30
  - 8.6% with bmi >30



Monthly Vaginal Ring

- **Ethinyl estradiol (EE) and Etonogestrel (ENG)**
- Worn for 3 weeks with a 1 week interval
- Low EE dose
  - 15 mcg EE
  - 120 mcg etonogestrel daily
- Continuous dosing
- No first-pass metabolism
  - No nausea
- Less break-through bleeding
- Failure rates:
  - Typical: 7%
  - Perfect: 0.3%

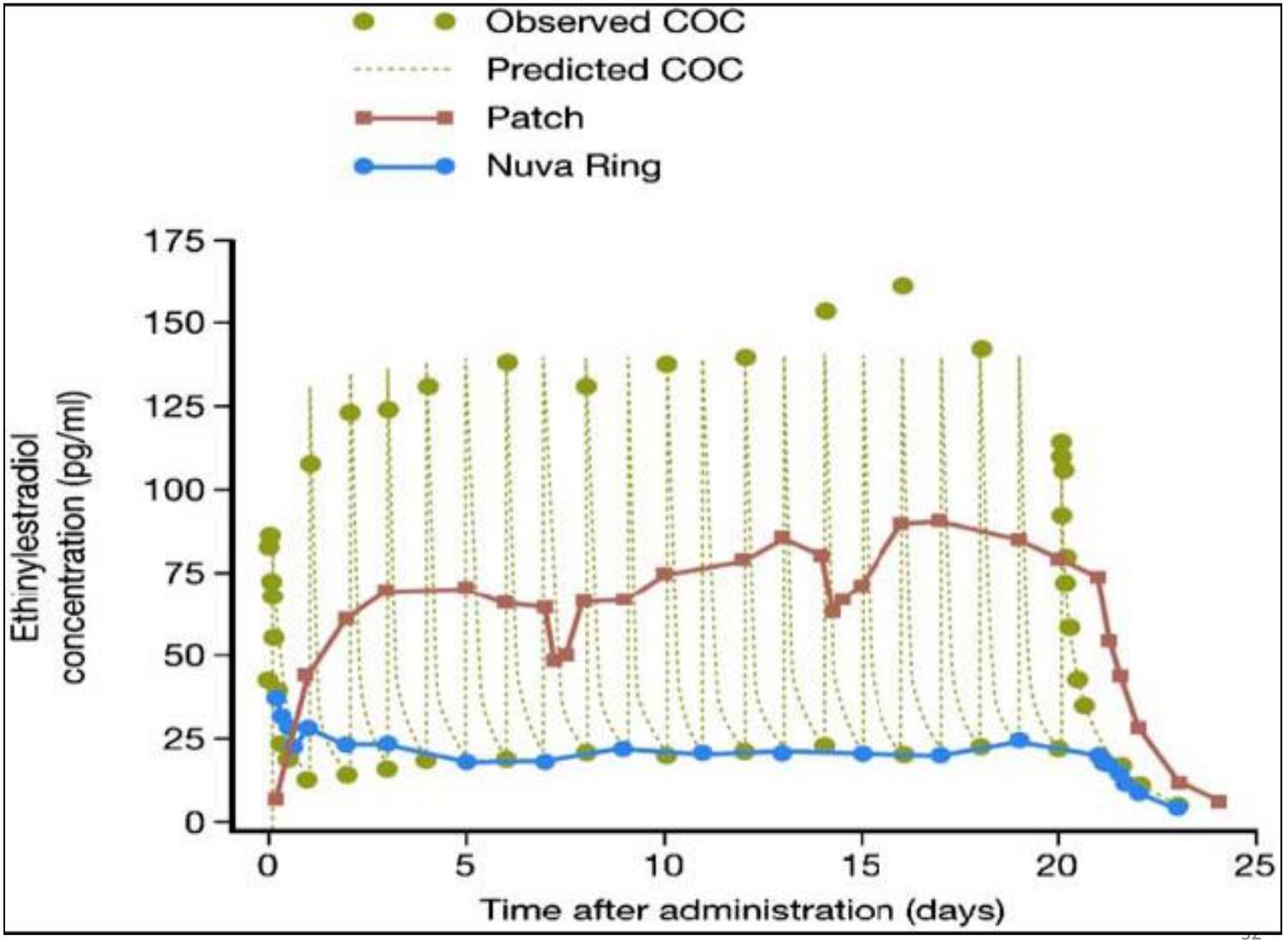


1 Year Vaginal Ring



- **Ethinyl estradiol (EE) and segesterone acetate (SE)**
- 1 ring for a whole year of use
- 0.013mg of EE and 0.15mg of SA released per day
- Wear for 3 weeks, remove for one week
- Store ring for the ring free week
- Replace the same ring
- Minimal breakthrough bleeding (one day per cycle)
- Perfect use: 2.98 pregnancies per 100 years of use

Pharmacokinetics of combined hormonal methods



7) Estrogen Containing Contraception

Risks

- Rare but serious
- Cardiovascular/thrombotic events
  - Stroke
  - Hypertensive urgency/emergency
  - Myocardial infarction
  - Thrombotic events: PE, DVT
    - Risk decreases with duration of use and decreasing estrogen dose

Contraindications

- History of thrombotic event
- Thrombotic disorder
- Hypertension
- Migraines with aura
- Tobacco use over the age of 35

US MEC
US SPR

CDC

Why do estrogen containing CHC increase VTE risk

Relative Risk and incidence of VTE							
Population	Young Patients	Pregnant Patients	High dose COC	Low dose COC	Leiden carrier	Leiden and COC	Leiden homozygous
RR	1	6	6-10	2-3	6-8	10-15	80
Incidence/ 10,000	5-10	30-60	30-100	10-30	30-80	50-100	400-800

- Causes increase in procoagulant factors V, VIII, X and fibrinogen
- Anticoagulant factors and proteins decrease
- Increase in prothrombin fragment 1 + 2
- Increased d-dimer
- Changes in endogenous thrombin potential
- **Risk decreases by 3 months of use**

Potential side effects of exogenous hormones

Estrogen Excess

- Nausea, breast tenderness, headaches, fluid retention
- Dysmenorrhea, menorrhagia, uterine fibroid growth

Estrogen Deficiency

- Early cycle (days 1-9) breakthrough bleeding and spotting
- Absence of withdrawal bleeding

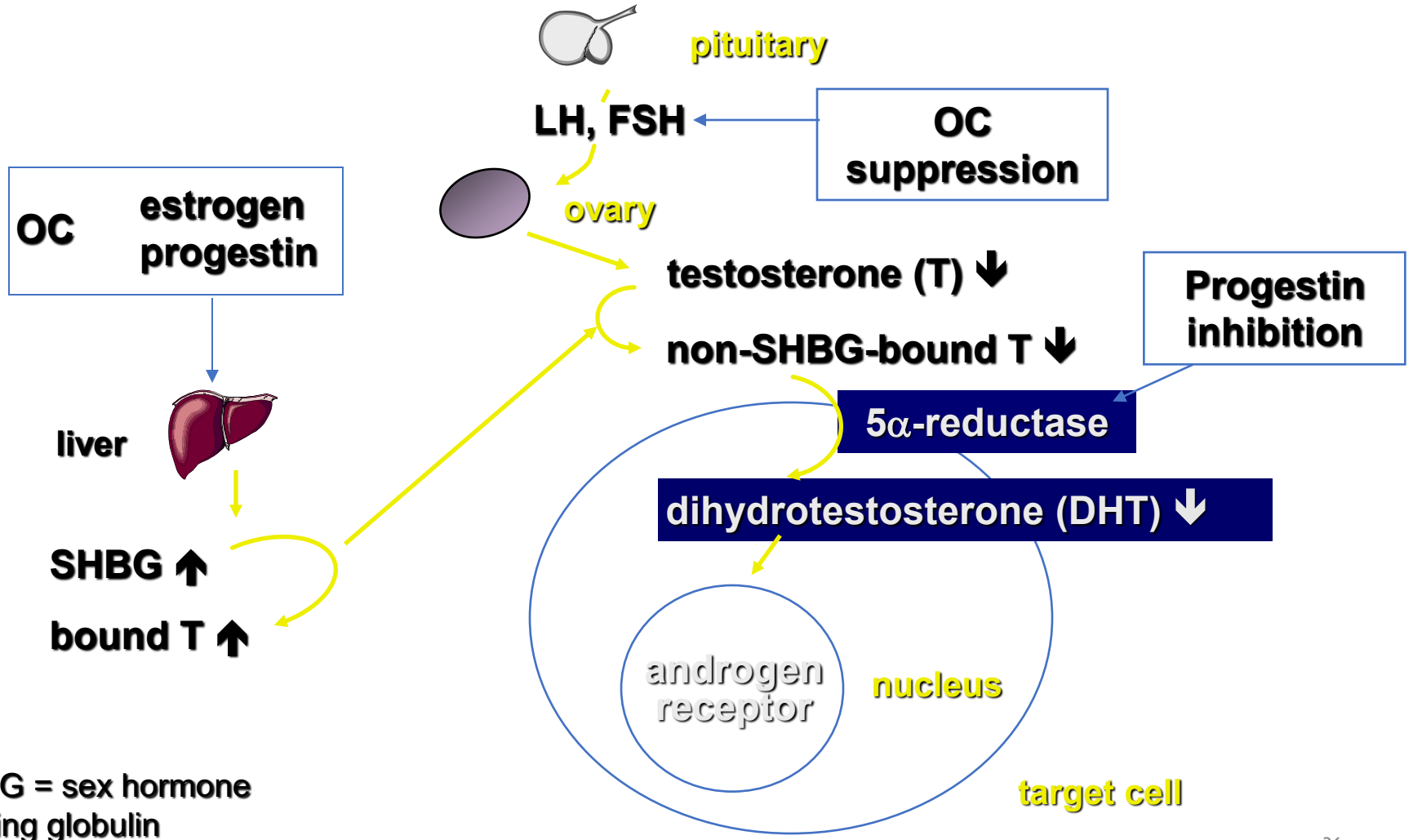
Progesterone Excess

- Increased appetite, weight gain
- Acne, oily skin, hirsutism
- Depression, fatigue, irritability

Progesterone Deficiency

- Heavy menses, dysmenorrhea
- Late-cycle (days 10-21) breakthrough bleeding/spotting

Effect of Combination OCs on Androgen Pathways



SHBG = sex hormone binding globulin

Effect of Combination OCs on Androgen Pathways

In summary – estrogen + progestin contraception result in:

- MORE sex hormone binding globulin → less free testosterone
- LESS testosterone synthesis → less free testosterone

FEWER androgenic effects (acne, hirsuitism, etc)

Clinical relevance

Androgenicity

- All CHC improve androgenic profile
- No randomized, controlled trial shows one low-dose OC is better treatment for acne or hirsutism

Weight gain

- No demonstrated association between CHC and weight changes

Depression

- No well demonstrated association between CHC and depression

8) Progestin-only contraception

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For it to work best, use it...	Every. Single. Day.	Every week	Every month	Every 3 months

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How does progestin contraception work?

Implant (Nexplanon)	- **Ovulation inhibition** - Cervical mucus thickening
Intrauterine device (Liletta, Mirena, Kyleena, Skyla)	- **Cervical mucus thickening**
Progestin-only pill (micronor, Opill, Slynd)	- **Cervical mucus thickening** - Drosperinone pill also suppresses ovulation
Injection (depo-provera)	- **Ovulation inhibition** - Cervical mucus thickening

Progestin-only Contraceptive Implant (Nexplanon)

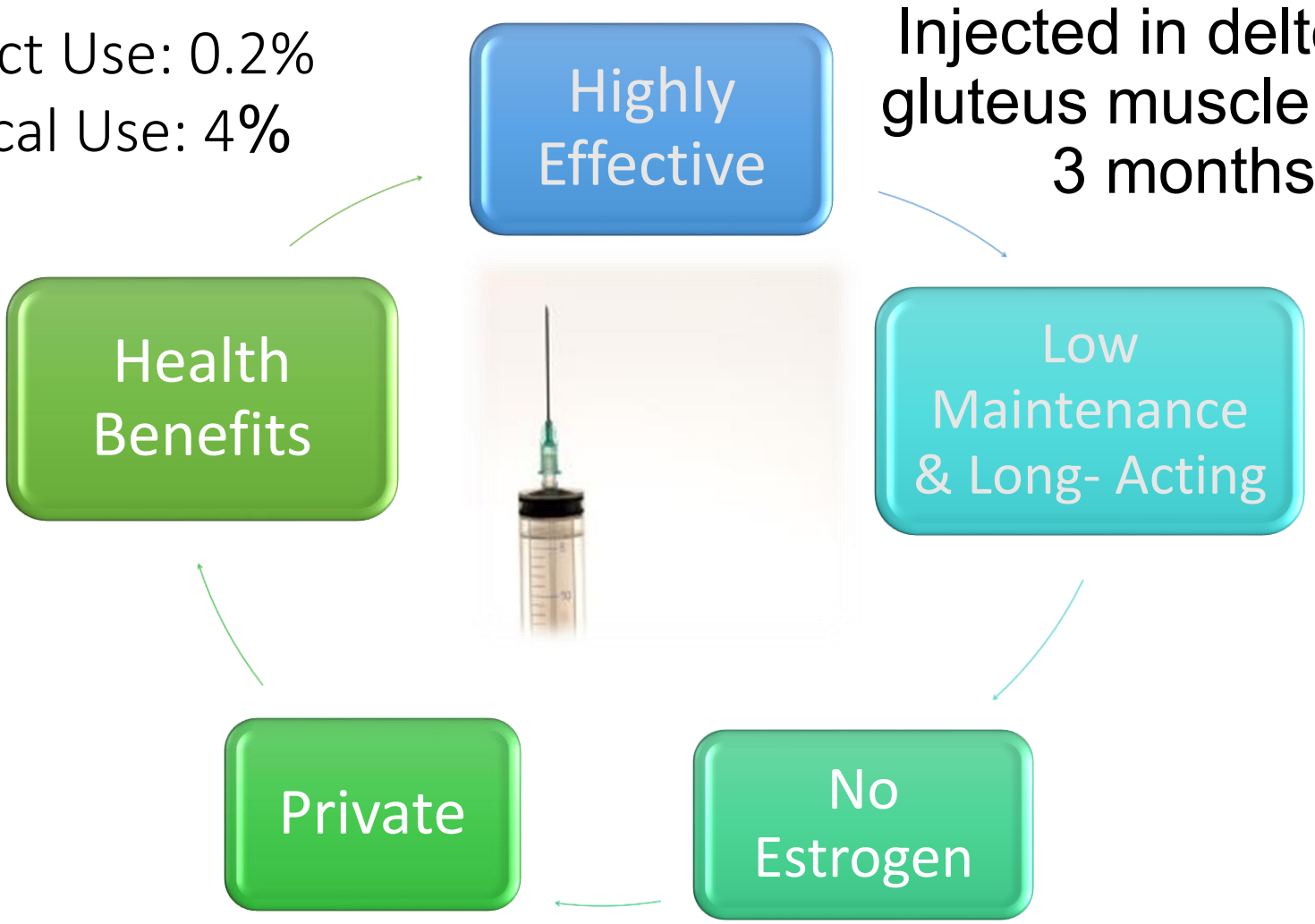
- Single 40-mm × 2-mm rod
  - Rod is made of ethylene vinyl acetate copolymer
- Contains 68 mg of etonogestrel
  - Releases 60 mcg progestin/day
- FDA approved for 3 years, effective for 5 years
- Failure rate: 0.01%
- Inhibits ovulation by altering GNRH release and suppressing the LH surge
- Progestin causes atrophy of endometrial glands
- Thickening of cervical mucus
- Side effect: menstrual changes



Injectable Contraception: Medroxyprogesterone Acetate (Depo-Provera, DMPA)

Perfect Use: 0.2%
Typical Use: 4%

Injected in deltoid or gluteus muscle every 3 months



DMPA side effects

- 1st several months: unpredictable or prolonged spotting
- After 1 year: 40-50% have amenorrhea
- 20-25% discontinuation because of menstrual issues
- Transient (reversible) bone mineral density loss
- **This is the only contraceptive associated with weight gain**

9) Cervical mucus thickeners



Progestin-only pills



Levonorgestrel Intrauterine devices

How do these thicken cervical mucus?

- The cervix is responsive to estrogen and progesterone
- Estrogen stimulates cervical mucus to be thin and penetrable by sperm
  - Estrogen levels are high at time of ovulation
  - Cervical mucus is thin at this time
- Progesterone thickens cervical mucus
  - Protective for pregnancy
- Steady progestin level ensures cervical mucus stays thick
  - Prevents sperm and possibly bacteria from entering upper genital tract

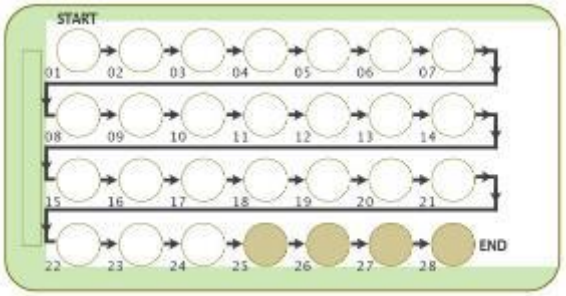
Progestin Only Pills (Norethindrone 0.35mg, Norgestrel 0.075mg)

- Can be taken by patients with contraindications to estrogen
  - Taken continuously with no hormone-free week
  - Must be taken at *exactly* the same time every day or decreased efficacy (3-hour window)
- Causes thickening of cervical mucus
  - Prevents sperm from traveling to upper genital tract
- Variable bleeding profile
- Failure rates: 3%



Progestin Only Pills (Drospirenone 4mg)





- Inhibits ovulation
  - Atrophy of endometrium
  - Thickens cervical mucus
- More forgiving dosing window
  - 24 hours window
- Spironolactone derived
  - Use with caution in those at risk for hyperkalemia

10) Hormonal Intrauterine Device: Levonorgestrel (LNG) IUDs

- Mechanism of action: Local progestin effect
  - Plasma concentration of progestin is 150-200pcg/dl
- Release progestin daily in varying doses
  - Causes thickening of cervical mucus
    - Prevents sperm from traveling to upper genital tract
  - Causes atrophy of uterine glands
    - Higher dose of progestin, higher the chance of amenorrhea
  - Does not inhibit ovulation
- Failure rate 0.1-0.4%

Comparing the 5 IUDs

IUD	*Lasts up to:*	*Total amount of hormones in device:*	
Liletta	Eight years	52 mg of levonorgestrel	T frame, no silver ring, blue threads
Skyla	Three years	13.5 mg of levonorgestrel	T frame, silver ring, grey threads
Kyleena	Five years	19.5 mg of levonorgestrel	T frame, silver ring, blue threads
Mirena	Eight years	52 mg of levonorgestrel	T frame, no silver ring, grey threads
ParaGard	Twelve years	None	T frame, copper wire, clear threads

11) Non-hormonal contraception

Non-Hormonal Contraception: Copper IUD

- Copper IUD (Paragard)
  - T-shaped IUD consisting of barium sulfate, polyethylene monofilament string and copper wire (380mm^2)
  - FDA Approved for 10 years, evidence supports 12 years
    - First year failure rate: 0.8%
    - 10-year cumulative failure rate: 2.1%
- Mechanism
  - Copper ions released into uterine environment
    - Sterile inflammatory reaction
    - Spermicidal
  - Can be used as EC after unprotected intercourse
- Advantages
  - No hormones
  - FDA approval for 10 years of contraception
- Disadvantages
  - Menses can be heavier and more cramping

Contraindications to IUD use

- Uterine anomalies, e.g. fibroids, congenital anomalies
- Blood dyscrasias
- Undiagnosed abnormal uterine bleeding
- Acute PID or history of PID with risk factor
- Untreated acute vaginitis or cervicitis
- Pregnancy or suspicion of pregnancy
- Carcinoma of cervix or endometrium
- Wilson’s disease or allergy to copper (for copper IUD)
- Genital actinomycosis

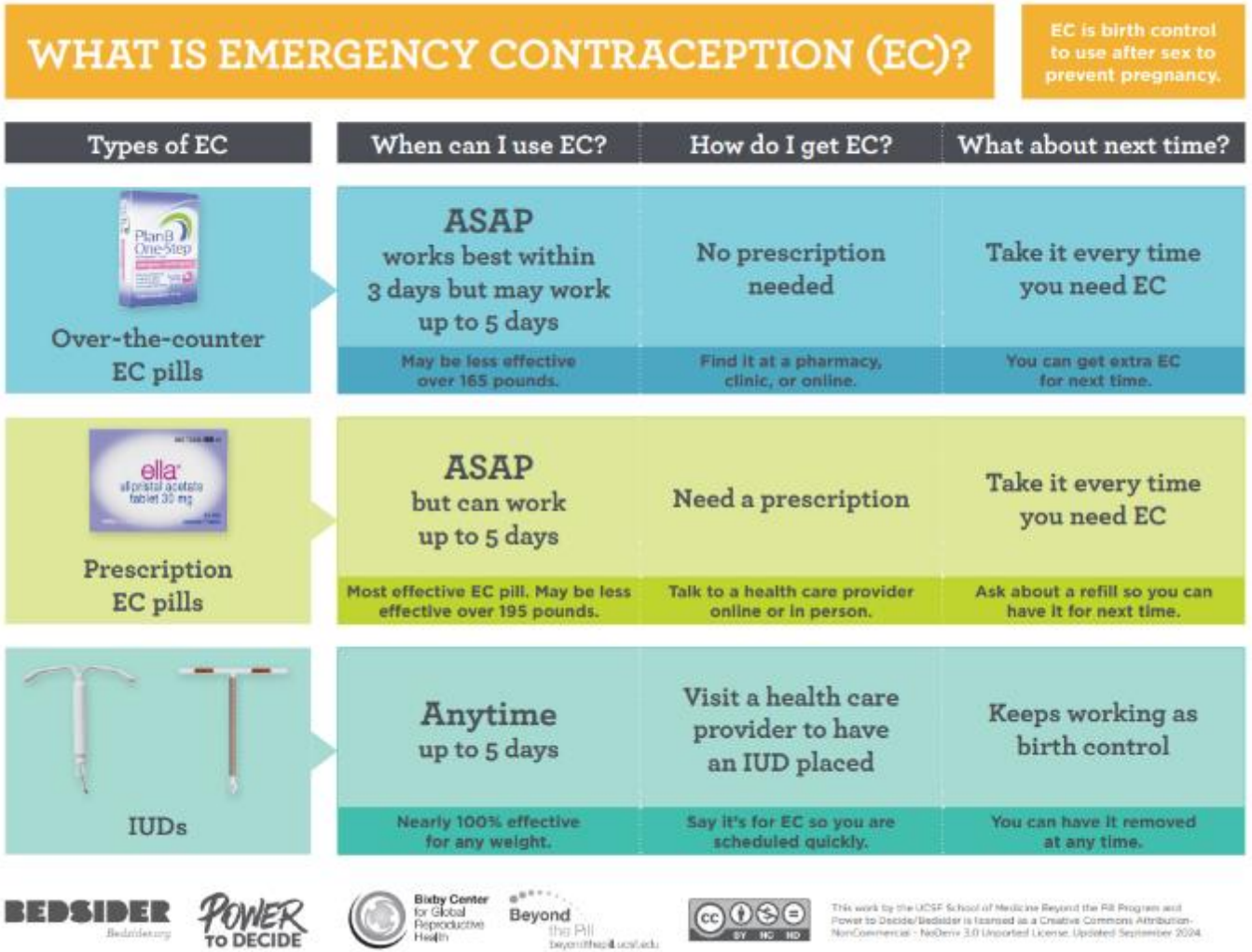
Vaginal Gel

- 1.8% lactic acid, 1% citric acid, 0.4% potassium bitartrate
- Keeps vaginal pH between 3.5 and 4.5
- Inhibits sperm motility
- Use up to one hour before each act of sex
- Comes in prefilled individually wrapped applicator



12) Emergency contraception

Post-coital contraception/ emergency contraception (EC)

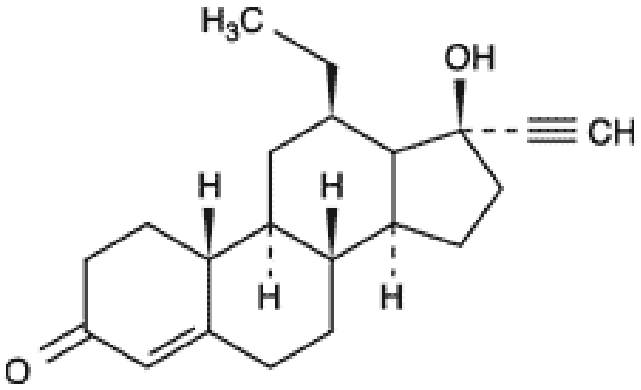


How does EC work?

- **Inhibits ovulation** by preventing LH surge (oral EC)
- **Inhibits sperm/tubal motility**, preventing fertilization (IUD)
- Does *not* appear to inhibit implantation
- Does *not* interrupt or affect an implanted, established pregnancy
- Does *not* have any serious side effects or contraindications, excluding allergy

What is Plan B?

- Levonorgestrel 1.5 mg x 1 dose
- Can be used up to 3 days after intercourse
  - More effective if taken ASAP
- Prevents LH surge
- Available without a prescription
- Less effective if BMI >30

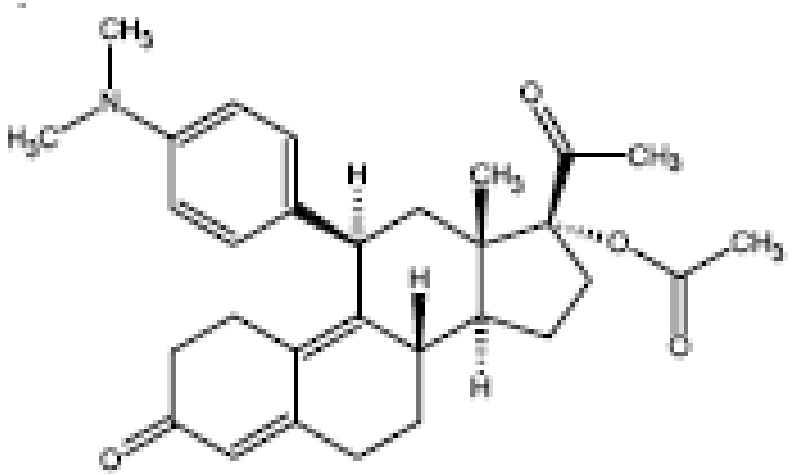




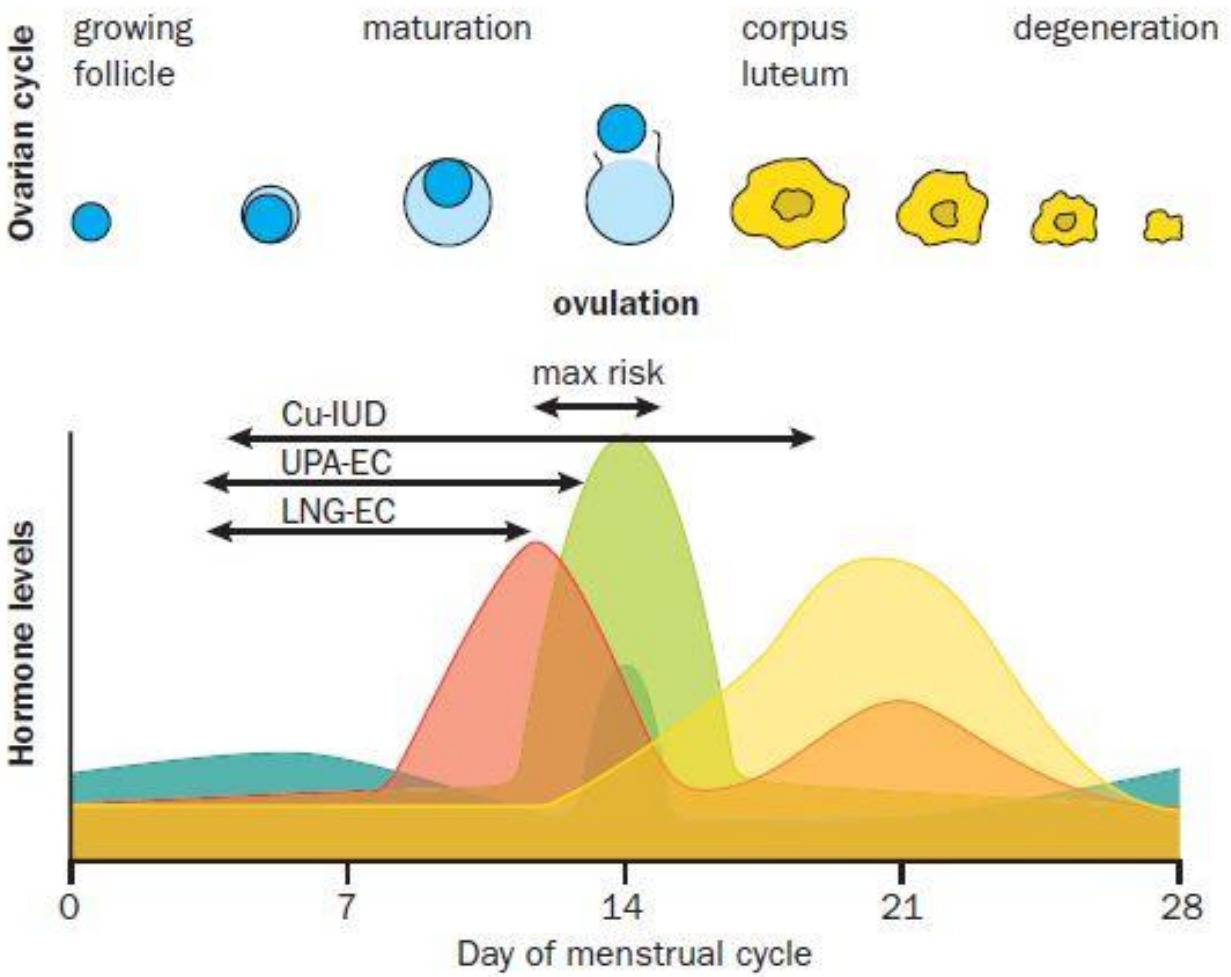
Available without prescription

What is ella®?

- Ulipristal: Selective progesterone receptor modulator
- One 30mg tablet. Rx only
- Can be used up to 5 days after intercourse
- Equally effective on days 1 and 5
- Less effective if BMI >35






Ovarian cycle
growing follicle
maturation
corpus luteum
degeneration
ovulation
max risk
Cu-IUD
UPA-EC
LNG-EC
Hormone levels
0
7
14
21
28
Day of menstrual cycle
Hormone:
progesterone
estrogen
luteinising hormone (LH)
follicle-stimulating hormone (FSH)

How well does EC work?

If 1000 people have unprotected intercourse in the 2nd or 3rd week of their menstrual cycle

- 80 (8%) will become pregnant without contraception
- 20 (2%) will become pregnant after use of the Yuzpe regimen (75% reduction)
- 10 (1%) will become pregnant after use of Plan B® (88% reduction)
- 1 (0.1%) will become pregnant after copper IUD insertion (99.9% reduction)

13) Medication abortion/Medical Management of Miscarriage



medication

- Avoids invasive procedure
- Available up to 10 weeks
- Occurs at home
- Multi step process
- Takes hours to days
- Active patient participation
- Requires follow-up



procedural

- Involves invasive procedure
- Available thru 2nd trimester
- Occurs in clinic
- Single step process
- Takes minutes
- Passive patient participation
- No follow-up required

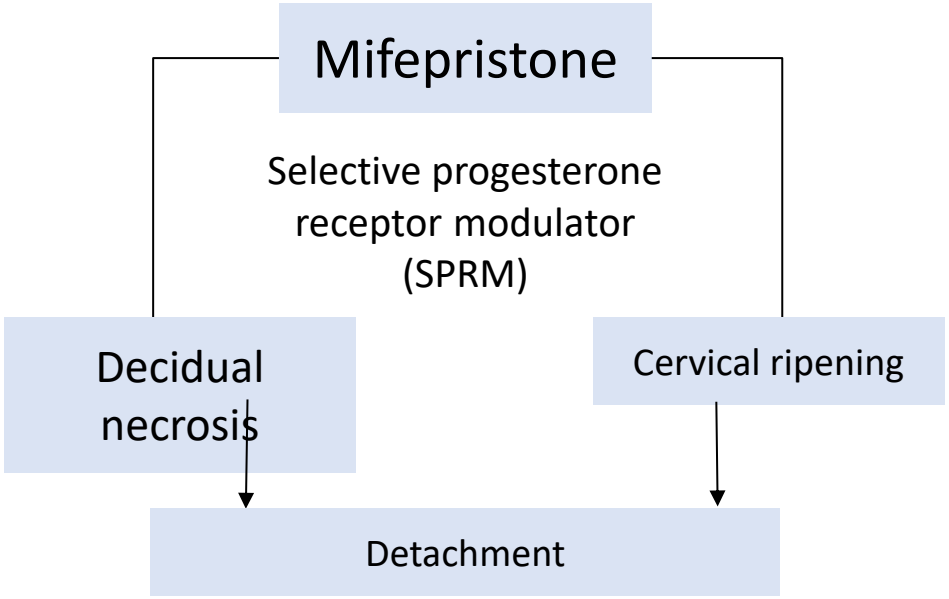
Medication Abortion (MAB)

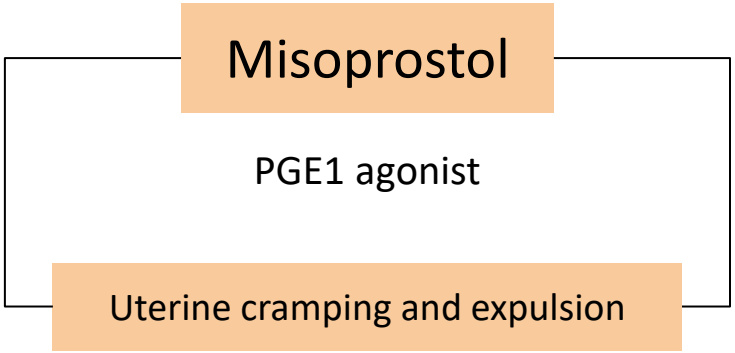
Table 1. Medication Abortion Regimens Up to 70 Days of Gestation

Regimen	Mifepristone Dose	Misoprostol Dose	Interval Between Drugs
Preferred			
Combination, FDA-approved*	200 mg (orally)	800 micrograms (buccally)	24–48 h
Combination, WHO recommended†	200 mg (orally)	800 micrograms (vaginally, sublingually, or buccally)	24–48 h
Alternative			
Misoprostol only	N/A	800 micrograms (vaginally, sublingually, or buccally)	Repeat every 3 h for up to 3 doses‡

ACOG Practice Bulletin #225

Medication Abortion (MAB)





Mifepristone Pharmacokinetics

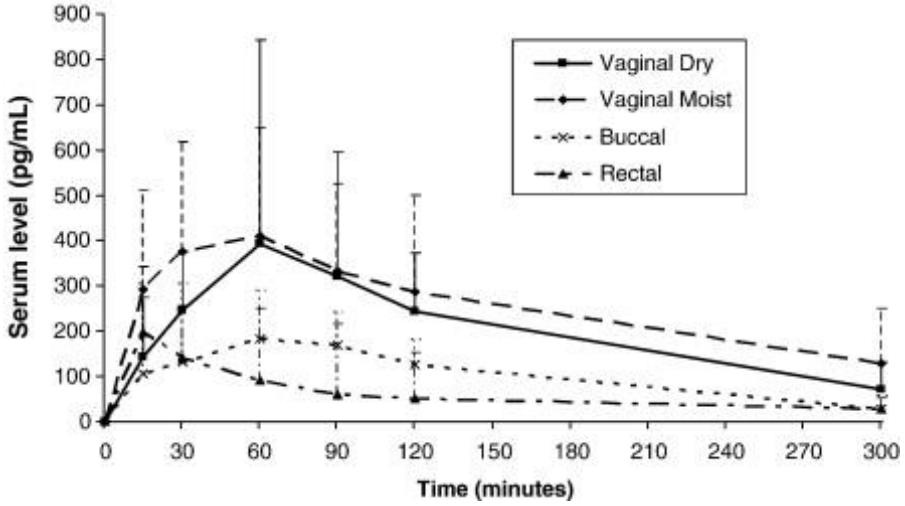
- Reaches peak serum concentrations in 2 hours
- At doses ≥100mg serum concentrations do not differ
  - Transport protein for mifepristone is saturated above 100mg
- It is likely that doses of 100mg are as efficacious as higher doses
- The half life of the drug is 24-29 hours

Heikenheimo et al. Contraception 1986; 34:613-24

Philibert et al. Endocrine Society Abstract 2006

Misoprostol

- Prostaglandin E analogue
- FDA approved to prevent/treat gastric ulcers
- Inexpensive
- Stable at room temperature
- Multiple routes of administration



Meckstroth, *et al*. Obstet Gynecol 2006; 108(3 pt 1):582-590.

FDA Regimen

- Ultrasound diagnosis of gestation ≤70 days
- Mifepristone 200mg orally
- Misoprostol 800mcg
  - Buccally 24-48 hours after mifepristone
- Combination causes expulsion of the pregnancy

- Need to confirm expulsion after treatment
  - Most commonly with ultrasound within 2 weeks
- Alternative evidence-based follow-up
  - Serial BHCG levels
  - Telephone follow-up with high sensitivity urine pregnancy testing

Contraindications

- Anemia (<9.0 g/dL Hgb)
- Anticoagulation/bleeding disorder
- Chronic steroid use/adrenal failure
- Porphyria



Thank You!

Questions?

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